



PSO 600 COMMISSIONING GUIDELINES

INDEX

TOPICS	PAGE
1. <i>Overview and Features</i>	3
2. <i>Feature Code List</i>	4
3. <i>Fault Codes list</i>	9
4. <i>Operator panel</i>	11
5. <i>Configuration mode</i>	13
6. <i>Terminal details</i>	15
7. <i>Automatic transfer system (AMF)</i>	18
8. <i>Meter calibration</i>	19
9. <i>Wiring diagram</i>	21
10. <i>Commissioning Report</i>	22

1.PSO 600 Overview & Features:

Overview:

The PS0600 controller is a microprocessor-based generator set monitoring, protection and START / STOP control system with AMF (Auto Mains Failure) feature. The control provides a simple operator interface to the alternator and engine parameters, remote and local start / stop control and generator set protective functions. The control does not have in-built speed governing and voltage regulation features. It supports magnetic pickup function for speed measurement if trim Magnetic pickup enabled.

The PS0600 controller is designed for use on Generating Sets from 10 to 125 kVA range, for non-paralleling applications. The control can also be configured for various frequency (50 Hz / 60 Hz operation), voltage configuration up to 600 Vrms L-L. The control is designed for mounting on the generator set. The PS0600 control is usually powered from the generator set starting batteries and works over a voltage range from 12 VDC to 24 VDC.

The control also has a 128x64 alphanumeric graphic LCD display and six tactile feel pushbuttons, which can be used for navigation.

Control Features:

- Operates on 12 VDC and 24 VDC
- RS-485 Communication for PC tool and Modbus RTU slave protocol
- Low power sleep mode
- Phase voltage and current sensing. Current sensing with an external 0 - 5 amp current transformers.
- Engine starting. Relay drivers (capable of driving 1A load absolute max) for start, Fuel Shut Off, Customer output 1 & 2. Start disconnect is achieved by monitoring the battery charging alternator voltage or main alternator frequency. There are no on-board relays present. External relays are required to be wired for all outputs
- Generator set monitoring. Displays critical engine and alternator parameters
- Auto mains failure detection and load transfer with open transition (Break before make).
- User settable Exercise scheduler to keep generator parts lubricating if not running for long duration
- Real time clock for events and logs

2.Feature Code list:

7.5kVA - 1Ph	10kVA - 1ph	15kVA - 1Ph	7.5kVA - 3Ph	10kVA - 3ph	15kVA - 3Ph	Description
A057Y864	A057Y864	A057Y864	A057Y864	A057Y864	A057Y864	GENSET APPLICATION RATING: PRIME
0326-6975	0326-6975	0326-6975	-	-	-	CODE-DES FEATURE COMMERCIAL COMMON (GENSET NOMINAL VOLTAGE:230V)
-	-	-	0326-6979	0326-6979	0326-6979	CODE-DES FEATURE COMMERCIAL COMMON (GENSET NOMINAL VOLTAGE:416V)
-	-	A055Y870	-	-	-	CODE-DES FEATURE PS0500 (CT RATIO:200:X)
-	-	-	A055Y886	A055Y886	-	CODE-DES FEATURE PS0500 (CT RATIO:50:X)
A055Y871	A055Y871	A055Y871	A055Y871	A055Y871	A055Y871	CODE-DES FEATURE COMMERCIAL COMMON (FREQUENCY SETTINGS: 50HZ)
A055Y914	A055Y914	A055Y914	N/A	N/A	N/A	CODE-DES FEATURE COMMERCIAL COMMON (SINGLE PHASE GENSET CONNECTION - 2 WIRE)
A056R615	A056R615	A056R615	A056R615	A056R615	A056R615	CODE-DES FEATURE COMMERCIAL COMMON (WAKE IN AUTO DISABLE)
A055Y869	A055Y869	-	-	-	A055Y869	CODE-DES FEATURE PS0500 (CT RATIO:100:X)
A055Y911	A055Y911	A055Y911	A055Y911	A055Y911	A055Y911	CODE-DES FEATURE PS0500 (OIL PRESSURE SENSOR CPGF(2-WIRE))
A055Y915	A055Y915	A055Y915	A055Y915	A055Y915	A055Y915	CODE-DES FEATURE PS0500 (CONFIG I/P#1:LOW FUEL WARNING)
A055Y935	A055Y935	A055Y935	A055Y935	A055Y935	A055Y935	CODE-DES FEATURE COMMERCIAL COMMON (X-SERIES CRANK CYCLE TIME)
A055Z021	A055Z021	A055Z021	A055Z021	A055Z021	A055Z021	CODE-DES FEATURE COMMERCIAL COMMON (Engine Protection Setpoints for X1.3)
A055Y972	A055Y972	A055Y972	A055Y972	A055Y972	A055Y972	CODE-DES FEATURE COMMERCIAL COMMON (12V Charging Alternator Disconnect Voltage:12 Volts)
A055Y933	-	-	A055Y933	-	-	CODE-DES FEATURE COMMERCIAL COMMON ((6 PRIME/6.6 STANDBY) kW 3P & 1P 50Hz) - 7.5KVA
-	A055Y934	-	-	A055Y934	-	CODE-DES FEATURE COMMERCIAL COMMON ((8 PRIME/8.8 STANDBY) kW 3P & 1P 50Hz) - 10kVA
-	-	A055Y991	-	-	A055Y991	CODE-DES FEATURE COMMERCIAL COMMON ((12 PRIME/13.2 STANDBY) kW 3P & 1P 50Hz) - 15kVA

X2.7G1	X2.7G2	X2.7G2	X2.7G1	X2.7G2	X2.7G2	
20kVA - 1Ph	25kVA - 1ph	30kVA - 1Ph	20kVA - 3Ph	25kVA - 3ph	30kVA - 3Ph	Description
A057Y864	A057Y864	A057Y864	A057Y864	A057Y864	A057Y864	GENSET APPLICATION RATING: PRIME
0326-6975	0326-6975	0326-6975	-	-	-	CODE-DES FEATURE COMMERCIAL COMMON (GENSET NOMINAL VOLTAGE: 230V)
-	-	-	0326-6979	0326-6979	0326-6979	CODE-DES FEATURE COMMERCIAL COMMON (GENSET NOMINAL VOLTAGE:416V)
-	A055Y964	A055Y964	-	-	-	CODE-DES FEATURE COMMERCIAL COMMON (CT RATIO:350:X)
-	-	-	A055Y869	A055Y869	A055Y869	CODE-DES FEATURE COMMERCIAL COMMON (CT RATIO:100:X)
A055Y871	A055Y871	A055Y871	A055Y871	A055Y871	A055Y871	CODE-DES FEATURE COMMERCIAL COMMON (FREQUENCY SETTINGS: 50HZ)
A055Y914	A055Y914	A055Y914	-	-	-	CODE-DES FEATURE COMMERCIAL COMMON (SINGLE PHASE GENSET CONNECTION - 2 WIRE)
A056R615	A056R615	A056R615	A056R615	A056R615	A056R615	CODE-DES FEATURE COMMERCIAL COMMON (WAKE IN AUTO DISABLE)
A055Y911	A055Y911	A055Y911	A055Y911	A055Y911	A055Y911	CODE-DES FEATURE PS0500 (OIL PRESSURE SENSOR CPGF(2-WIRE))
A055Y915	A055Y915	A055Y915	A055Y915	A055Y915	A055Y915	CODE-DES FEATURE PS0500 (CONFIG I/P#1:LOW FUEL WARNING)
A055Y870	-	-	-	-	-	CODE-DES FEATURE COMMERCIAL COMMON (CT RATIO:200:X)
A055Z020	A055Z020	A055Z020	A055Z020	A055Z020	A055Z020	CODE-DES FEATURE COMMERCIAL COMMON (Engine Protection Setpoints for X2.7 and X3.6)
A055Y972	A055Y972	A055Y972	A055Y972	A055Y972	A055Y972	CODE-DES FEATURE COMMERCIAL COMMON (12V Charging Alternator Disconnect Voltage:12 Volts)
A055Y992	-	-	A055Y992	-	-	CODE-DES FEATURE COMMERCIAL COMMON ((16 PRIME/17.6 STANDBY) kW 3P & 1P 50Hz) - 20kVA
-	A055Y993	-	-	A055Y993	-	CODE-DES FEATURE COMMERCIAL COMMON ((20 PRIME/22 STANDBY) kW 3P & 1P 50Hz) - 25kVA
-	-	A055Y994	-	-	A055Y994	CODE-DES FEATURE COMMERCIAL COMMON ((24 PRIME/26.4 STANDBY) kW 3P & 1P 50Hz) - 30kVA
	-	-	-	-	-	CODE-DES FEATURE COMMERCIAL COMMON 1P 50Hz - 27kVA

X3.6G1	X3.6G1	X3.6G1	
35kVA - 3Ph	40kVA - 3ph	35kVA - 1Ph	Description
A057Y864	A057Y864	A057Y864	GENSET APPLICATION RATING: PRIME
0326-6979	0326-6979	-	CODE-DES FEATURE COMMERCIAL COMMON (GENSET NOMINAL VOLTAGE: 415V)
-	-	0326-6975	CODE-DES FEATURE COMMERCIAL COMMON (GENSET NOMINAL VOLTAGE: 230V)
A055Y871	A055Y871	A055Y871	CODE-DES FEATURE COMMERCIAL COMMON (FREQUENCY SETTINGS: 50HZ)
A056R615	A056R615	A056R615	CODE-DES FEATURE COMMERCIAL COMMON (WAKE IN AUTO DISABLE)
A055Y911	A055Y911	A055Y911	CODE-DES FEATURE PS0500 (OIL PRESSURE SENSOR CPGF(2-WIRE))
A055Y870	A055Y870	-	CODE-DES FEATURE COMMERCIAL COMMON (CT RATIO:200:X)
-	-	A055Y903	CODE-DES FEATURE COMMERCIAL COMMON (CT RATIO:300:X)
A055Y972	A055Y972	A055Y972	CODE-DES FEATURE COMMERCIAL COMMON (12V Charging Alternator Disconnect Voltage:12 Volts)
N/A	N/A	N/A	CODE-DES FEATURE PS0500 (Cold Start Advance Solenoid Temperature Function Enable - Enabled)
A055Z020	A055Z020	A055Z020	CODE-DES FEATURE COMMERCIAL COMMON (Engine Protection Setpoints for X2.7 and X3.6)
N/A	N/A	N/A	CODE-DES FEATURE PS0500 (Configurable Customer Output 1 - 5671)
A055Y999	-	A055Y999	CODE-DES FEATURE COMMERCIAL COMMON (28 PRIME/30.8 STANDBY kW 3P 50Hz) - 35kVA
-	A055Z001	-	CODE-DES FEATURE COMMERCIAL COMMON (32 PRIME/35.2 STANDBY kW 3P 50Hz) - 40kVA

50kVA - 3Ph	62.5kVA - 3ph	50kVA - 1Ph	62.5kVA - 1ph	Description
A057Y864	A057Y864	A057Y864	A057Y864	GENSET APPLICATION RATING: PRIME
0326-6979	0326-6979	-	-	CODE-DES FEATURE COMMERCIAL COMMON (GENSET NOMINAL VOLTAGE: 416V)
-	-	0326-6975	0326-6975	CODE-DES FEATURE COMMERCIAL COMMON (GENSET NOMINAL VOLTAGE: 230V)
-	A055Y903	-	-	CODE-DES FEATURE COMMERCIAL COMMON (CT RATIO:300:X)
-	-	A055Y966	-	CODE-DES FEATURE COMMERCIAL COMMON (CT RATIO:500:X)
-	-	-	XXXXX	CODE-DES FEATURE COMMERCIAL COMMON (CT RATIO:750:X)
A055Y871	A055Y871	A055Y871	A055Y871	CODE-DES FEATURE COMMERCIAL COMMON (FREQUENCY SETTINGS: 50HZ)
A056R615	A056R615	A056R615	A056R615	CODE-DES FEATURE COMMERCIAL COMMON (WAKE IN AUTO DISABLE)
A055Y911	A055Y911	A055Y911	A055Y911	CODE-DES FEATURE PS0500 (OIL PRESSURE SENSOR CPGF(2-WIRE))
A055Y870	-	-	-	CODE-DES FEATURE COMMERCIAL COMMON (CT RATIO:200:X)
A055Y983	A055Y983	A055Y983	A055Y983	CODE-DES FEATURE COMMERCIAL COMMON (LOP Shutdown Threshold (22 psi) and LOP Warning Threshold (32 psi))
A055Y972	A055Y972	A055Y972	A055Y972	CODE-DES FEATURE COMMERCIAL COMMON (12V Charging Alternator Disconnect Voltage:12 Volts)
A055Y995	-	A055Y995	-	CODE-DES FEATURE COMMERCIAL COMMON ((40 PRIME/44 STANDBY) kW 3P & 1P 50Hz) - 50kVA
-	A055Y998	-	A055Y998	CODE-DES FEATURE COMMERCIAL COMMON (50 PRIME/55 STANDBY kW 3P & 1P 50Hz) - 62.5kVA

100kVA - 3Ph	125kVA - 3ph	Description
A057Y864	A057Y864	GENSET APPLICATION RATING: PRIME
0326-6979	0326-6979	CODE-DES FEATURE COMMERCIAL COMMON (GENSET NOMINAL VOLTAGE: 416V)
A055Y871	A055Y871	CODE-DES FEATURE COMMERCIAL COMMON (FREQUENCY SETTINGS: 50HZ)
A056R615	A056R615	CODE-DES FEATURE COMMERCIAL COMMON (WAKE IN AUTO DISABLE)
A055Y911	A055Y911	CODE-DES FEATURE PS0500 (OIL PRESSURE SENSOR CPGF(2-WIRE))
A055Y966	A055Y966	CODE-DES FEATURE COMMERCIAL COMMON (CT RATIO:500:X)
A055Y970	A055Y970	HCT PROTECTION
A055Y983	A055Y983	CODE-DES FEATURE COMMERCIAL COMMON (LOP Shutdown Threshold (22 psi) and LOP Warning Threshold (32 psi))
A055Y972	A055Y972	CODE-DES FEATURE COMMERCIAL COMMON (12V Charging Alternator Disconnect Voltage:11 Volts)
A055Z003	-	CODE-DES FEATURE COMMERCIAL COMMON (80 PRIME/88 STANDBY kW 3P 50Hz) - 100kVA
-	A055Z004	CODE-DES FEATURE COMMERCIAL COMMON (100 PRIME/110 STANDBY kW 3P 50Hz) - 125kVA
A056L544	A056L544	CODE-DES FEATURE COMMERCIAL COMMON (LCT warning threshold 41DegF)

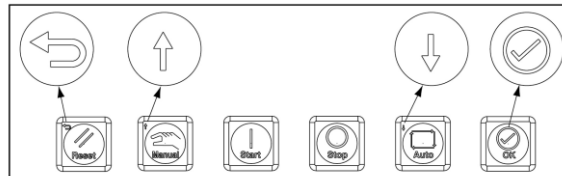
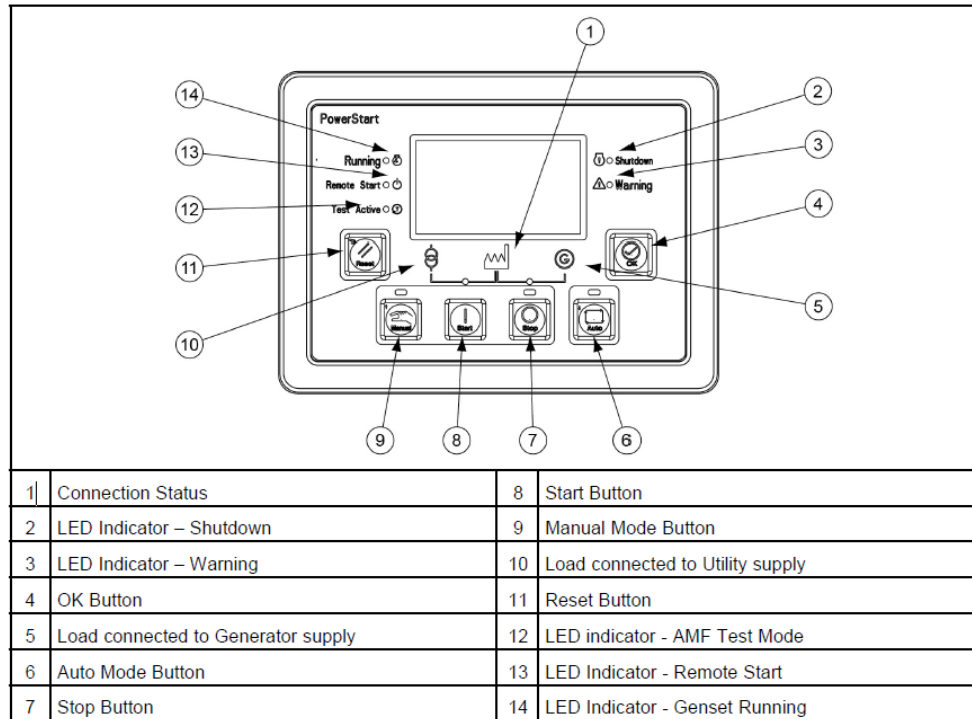
70kVA - 3Ph	82.5kVA - 3ph	Description
A057Y864	A057Y864	GENSET APPLICATION RATING: PRIME
0326-6979	0326-6979	CODE-DES FEATURE COMMERCIAL COMMON (GENSET NOMINAL VOLTAGE: 416V)
A055Y871	A055Y871	CODE-DES FEATURE COMMERCIAL COMMON (FREQUENCY SETTINGS: 50HZ)
A056R615	A056R615	CODE-DES FEATURE COMMERCIAL COMMON (WAKE IN AUTO DISABLE)
A055Y903	A055Y903	CODE-DES FEATURE COMMERCIAL COMMON (CT RATIO:300:X)
A055Y911	A055Y911	CODE-DES FEATURE PS0500 (OIL PRESSURE SENSOR CPGF(2-WIRE))
A055Y970	A055Y970	HCT PROTECTION
A055Y983	A055Y983	CODE-DES FEATURE COMMERCIAL COMMON (LOP Shutdown Threshold (22 psi) and LOP Warning Threshold (32 psi))
A055Y972	A055Y972	CODE-DES FEATURE COMMERCIAL COMMON (12V Charging Alternator Disconnect Voltage:12 Volts)
A055Z000	-	CODE-DES FEATURE COMMERCIAL COMMON (56 PRIME/61.6 STANDBY kW 3P 50Hz) - 70KVA
-	A055Z002	CODE-DES FEATURE COMMERCIAL COMMON (66 PRIME/72.6 STANDBY kW 3P 50Hz) - 82.5kVA
A056L544	A056L544	CODE-DES FEATURE COMMERCIAL COMMON (LCT warning threshold 41DegF)

3.Fault Code List:

S. No	Fault Description	Fault Code
Shutdown Faults		
1	Loss of speed	115
2	High Coolant Temp	151
3	Overspeed	234
4	Fail To Start	359
5	Low Oil Pressure	415
6	Shutdown After Battle short	1123
7	Config Input #2 Fault	1312
8	Config Input #3 Fault	1317
9	Remote E-Stop	1434
10	Fail To Crank	1438
11	High AC Voltage	1446
12	Low AC Voltage	1447
13	Under frequency	1448
14	Over frequency	1449
15	Gen CB Fail To Close	1452
16	Speed/Hz Mismatch	1469
17	Over Current / High AC current	1472
18	Common warning	1540
19	Config Input #1 Fault	1573
20	Loss of AC sense	2335
21	Fail To Stop	2677
22	Genset CT Ratio Low	2814
23	Modbus Network Failure	4799
24	Genset Overload	5135

S.No	Fault Description	Fault Code
	Warning Faults	
1	Oil Pressure OORH	135
2	Oil Pressure OORL	141
3	Low Oil Pressure	143
4	Coolant Temperature OORH	144
5	Coolant Temperature OORL	145
6	High Coolant Temp	146
7	Low Battery Voltage	441
8	High Battery Voltage	442
9	Battle Short Active	1131
10	Utility Frequency Error	1223
11	Fail To Shutdown	1416
12	Power Down Failure	1417
13	Low Coolant Temperature	1435
14	Low Fuel Level	1441
15	Weak Battery	1442
16	kW Overload	1444
17	Genset CB Fail To Open	1453
18	Over Current	1471
19	Common shutdown	1541
20	Real Time Clock Reset	1689
21	Fuel Level OORH	2222
22	Fuel level OORL	2223
23	Utility Under voltage	2331
24	Utility Overvoltage	2358
25	Utility CB Fail To Close	2396
26	Utility CB Fail To Open	2397
27	Charging Alternator Failure	2678
28	Modbus Failure	2939
29	Maintenance Due	6226
30	Current Imbalance	6227

4.Operator Panel:



Symbol	Functionality
	In configuration mode following functions can be done: Navigates to previous screen Navigates to previous sub menu Navigates to parameter on same screen Increments parameter value (change enumeration)
	In configuration mode following functions can be done: Navigates to next screen Navigates to next submenu Navigates to parameter on same screen Decrements parameter value (change enumeration)
	Returns to previous screen Shall jump to Scrolling/Non-scrolling Mode from configuration mode
	Selects screen Selects parameter Shall change mode from Non-scrolling to scrolling and vice-versa Saves in non-volatile memory

Control Buttons:

Manual:

Press this button to put the control in Manual mode from Off mode. If the **Start** button is not pressed within five seconds of the pressing of the **Manual** Button, the control automatically returns to Off mode. The green LED above this button lights when the control is in Manual mode.

Auto:

Press this button to put the control in Auto mode from Off mode. The green LED above this button lights when the control is in Auto mode. In Auto mode, the generator set starts and runs when a remote start signal is received.

Start:

In Manual mode, press this button to initiate a Manual Start sequence. In Auto mode, this button has no effect.

Stop:

Press this button to switch to the Off mode from both Manual as well as Auto modes. The Red LED above this button lights when the control is in Off mode.

Reset:

This button wakes the control from Power Down Sleep Mode. The Controls enters in configuration mode when **Reset** and **OK** buttons are pressed together. To enter in configuration mode it is necessary that the control is in Stop mode.

OK:

This button is used to acknowledge the fault pop-ups. It is also used to Pause / Resume the scrolling of the parameter frames. It is used to select the menu items and the parameters in the selected menu in the configuration mode. The selected parameter is preceded by an asterisk (*) and will be saved when the **Auto** button is pressed to exit the configuration mode.

LED Indicators:

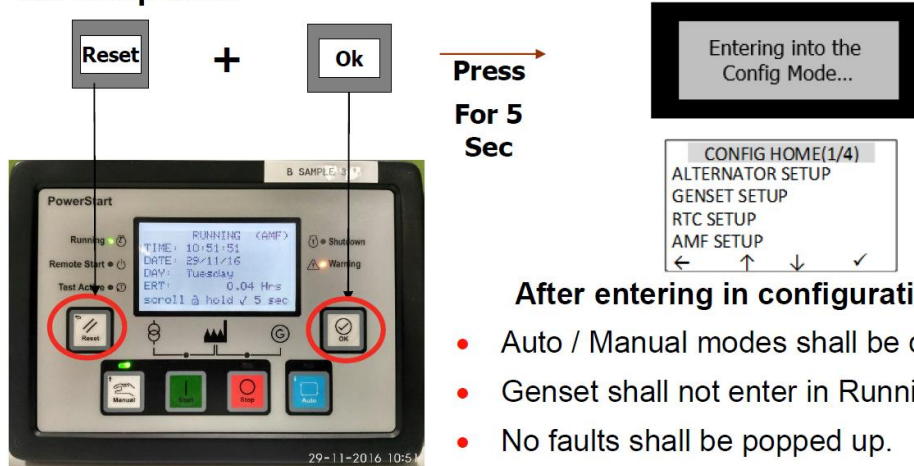
The operator panel shall have 10 LED indicators out of which 7 shall be status LEDs and the remaining 3 shall indicate the controller mode. Colours and respective functionality is included in the table below.

LED	Colour	Functionality
Generator Running	GREEN	Shall indicate that the Generator is in running state
Remote Start	GREEN	Shall indicate that a remote start command is active.
Test Active	GREEN	Shall indicate Auto Mains Failure Test mode is Active
Warning	AMBER	Shall indicate that there is an occurrence of a warning fault.
Generator Shutdown	RED	Shall indicate that there is an occurrence of a Generator shutdown fault.
Auto	GREEN	Shall indicate that the genset control is in Auto mode
Manual	GREEN	Shall indicate that the genset control is in Manual mode
Stop	RED	Shall indicate that the Genset is in OFF (Stop) Mode"
Connected to Genset	GREEN	Shall indicate load is connected to Genset
Connected to Utility	GREEN	Shall indicate load is connected to Utility

5.Configuration Mode

To enter into the Set up mode :

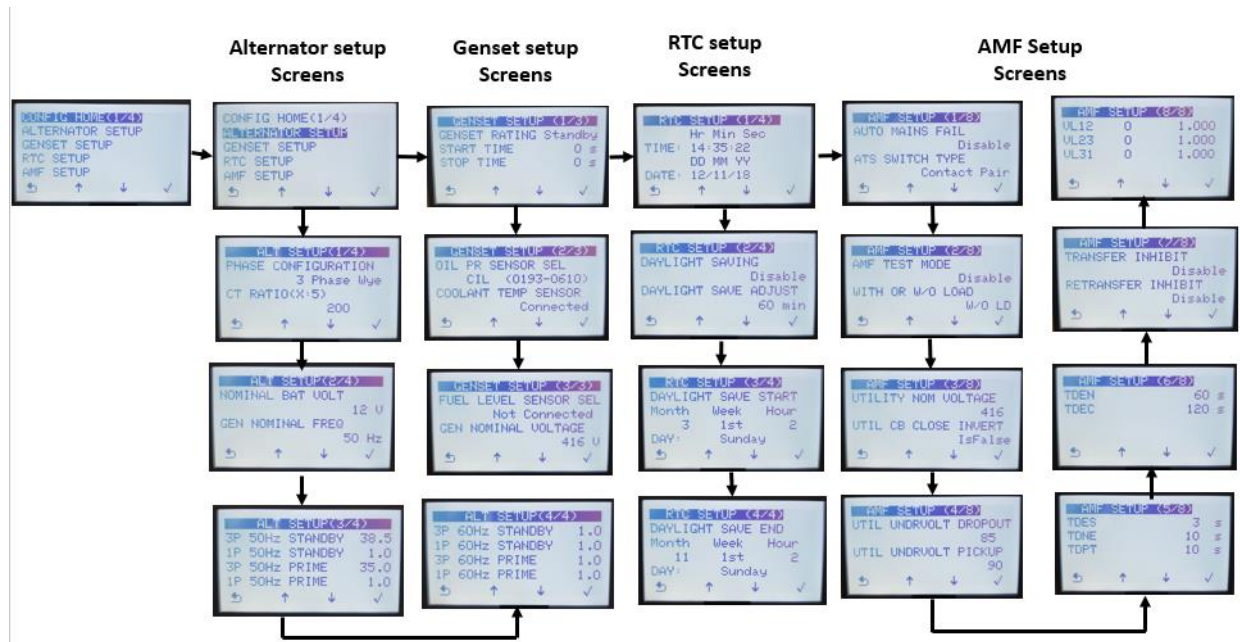
➤ After going to the stop mode only we can enter the Setup mode



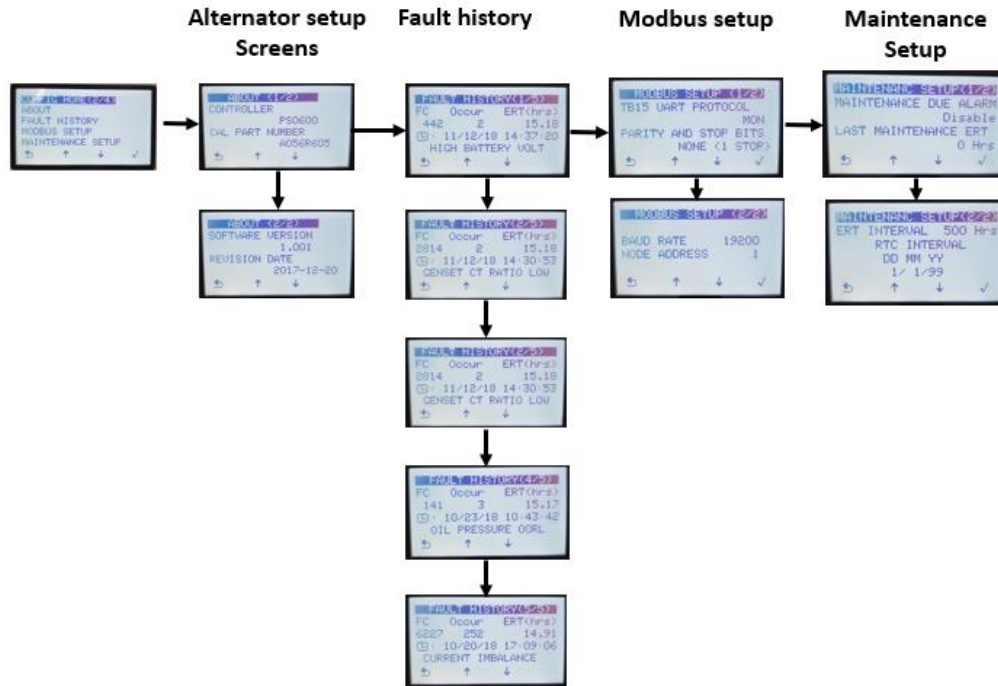
After entering in configuration mode:

- Auto / Manual modes shall be disabled.
- Genset shall not enter in Running state.
- No faults shall be popped up.

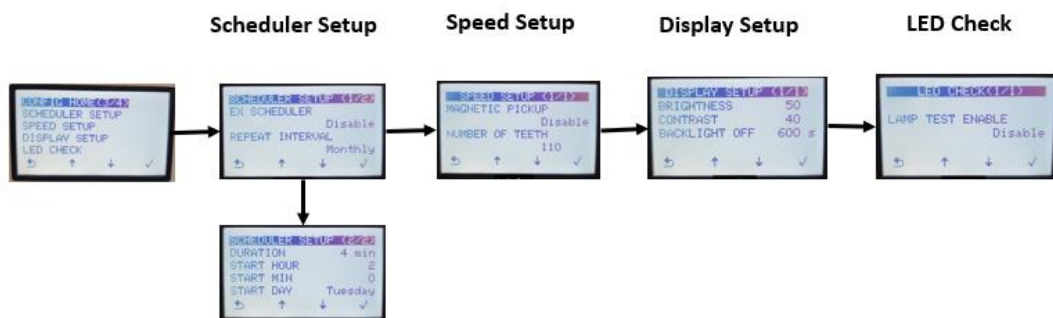
Configuration Home Screen-1:



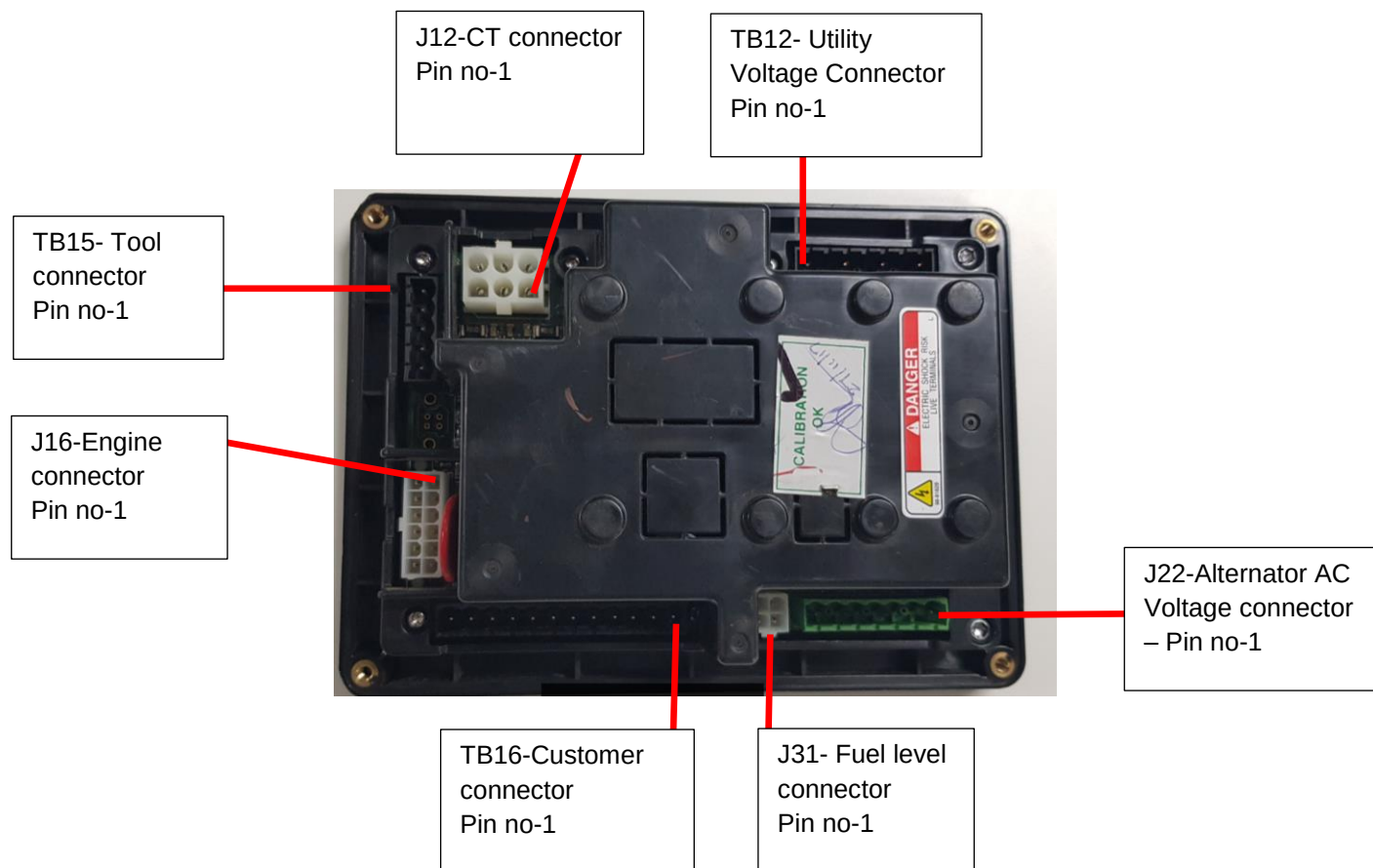
Configuration Home Screen-2:



Configuration Home Screen-3:



6.Terminal details



Engine Connector:

The engine connector (J16) is a 12-pin connector-plug (12 MINI M-N-L). Connector pin mapping details are as follows:

Connector Pin	Signal Name
J16-1	COOLANT TEMPERATURE SENSOR INPUT
J16-2	BATTERY CHARGER/ALTERNATOR WL CONNECTION
J16-3	STARTER OUTPUT (LOW SIDE)
J16-4	FUEL SHUTOFF HOLD COIL (LOW SIDE)
J16-5	BATTERY -VE
J16-6	CHASIS GND
J16-7	
J16-8	
J16-9	OIL PRESSURE SENDER INPUT
J16-10	STARTER DISCONNECT INPUT
J16-11	COMMON GND
J16-12	10 A FUSED BAT+

CT Connector:

The CT connector J12 is a 6-pin MATE-N-LOCK connector. Connector pin mapping details are as follows:

Connector pin	Signal
J12-1	CT1_INPUT
J12-2	CT2_INPUT
J12-3	CT3_INPUT
J12-4	CT1_COM
J12-5	CT2_COM
J12-6	CT3_COM

Genset Voltage Connector:

The voltage connection is via a 4-pin header with alternate spaces between the pins. The Generator voltage connector is J22. This connector is GREEN in color. Connector pin mapping details are as follows;

Connector pin	Signal
J22-1	GEN_V1
J22-2	GEN_V2
J22-3	GEN_V3
J22-4	GEN_N

Utility Voltage Connector:

The voltage connection is via a 4-pin header with alternate spaces between the pins. The utility voltage connector is TB7. This connector is BLACK in color. Connector pin mapping details are as follows:

Connector pin	Signal
TB7-1	UTIL_V1
TB7-2	UTIL_V2
TB7-3	UTIL_V3
TB7-4	UTIL_N

Customer Connection TB:

The customer connection TB16 is a 12-pin terminal block. Connector pin mapping details are as follows:

Terminal pins	Signal
TB16-1	CUSTOMER_INPUT_2 / UTILITY CB FEEDBACK
TB16-2	CUSTOMER_INPUT_3 / GENSET CB FEEDBACK
TB16-3	CUSTOMER_OP3 / UTILITY CB CLOSE LOW SIDE DRIVER
TB16-4	CUSTOMER_OP4 / UTILITY CB OPEN LOW SIDE DRIVER
TB16-5	CUSTOMER_OP5 / GENSET CB CLOSE LOW SIDE DRIVER
TB16-6	CUSTOMER_OP6 / GENSET CB OPEN LOW SIDE DRIVER
TB16-7	REMOTE_START
TB16-8	CUSTOMER_OP1
TB16-9	CUSTOMER_OP2
TB16-10	CUSTOMER_INPUT_1
TB16-11	ESTOP_1
TB16-12	NC

Returns for all the above connections are required to be connected externally.

All the inputs are active GND unless they are configured for Active Open. Remote E-Stop switch should have two independent potential free contacts. One NC contact for cutting out the Battery supply to the Fuel Solenoid and another NC contact to disconnect controller's E-Stop input from GND to indicate to the control when the E-Stop is pushed in active state.



Battery Positive is not provided on the Customer terminal block (TB16). External provision needs to be made in the harness to provide the B+ signal for convenience of wiring up external relays and may be used as relay supply (high side) for FSO, Starter, Customer Output and Glow plug Relays via a 1 Amp Fuse.

PC Tool / Modbus communication Connector:

The PC tool connector TB15 is a dedicated 5 pin connector header block. Connector pin mapping details are as follows:

Terminal pins	Signal
TB15-1	GND
TB15-2	NC
TB15-3	RS 485 DATA A
TB15-4	RS 485 DATA B
TB15-5	WAKEUP

Fuel level sensor input connector:

The Fuel level connector is mini fit junior connector J31 is a dedicated 4 pin connector header block. Connector pin mapping details are as follows:

Terminal pins	Signal
J31-1	NC
J31-2	Fused Bat +
J31-3	NC
J31-4	Fuel level sensor input (4-20mA)

7. Automatic Transfer System (ATS / AMF)

The Automatic Transfer System (ATS) function applies when the Auto Mains Failure (AMF) Enable (trim) is enabled. ATS primarily means that the genset controller is controlling both the genset breaker and a utility breaker in a transfer pair arrangement. ATS is only for use in a single genset / single utility arrangement. ATS's primary job is to keep loads powered. ATS completely manages the system by automatically starting the genset and transferring load when it detects utility failure. ATS has numerous built-in configurable sensors to determine the availability of the utility and genset sources. Sensors include under voltage, overvoltage, over/under frequency and breaker failure. ATS has adjustable timers as well, including programmed transition delay, transfer delay and retransfer delay. PSO600 control supports only open transition (Break before Make) ATS functionality.

The control needs following trims to be adjusted for ATS functionality:

1. Auto Mains Failure Enable = Enable
2. Load transfer switch type:

a. Contact Pair:

ATS switch type is default to 'Contact pair'. When ATS switch type is 'Contact pair' the Genset and Utility breaker commands are of continuous type. In this case only two digital outputs are used for CB activation:

Terminal Pins	Signal
TB16-1	Utility CB feedback
TB16-2	Genset CB feedback
TB16-3	Utility CB close command
TB16-4	CUSTOMER_OP4
TB16-5	Genset CB close command
TB16-6	CUSTOMER_OP6
TB16-7	REMOTE_START
TB16-8	CUSTOMER_OP1
TB16-9	CUSTOMER_OP2
TB16-10	CUSTOMER_INPUT_1
TB16-11	ESTOP_1
TB16-12	NC

b. GTEC:

If GTEC or ACB is used then the ATS switch type must be changed to GTEC. When ATS switch type is 'GTEC' the Genset and utility breaker commands are of pulse type. In this case open and close commands are used for both utility and Genset CB's. So there are 4 digital outputs activated by control.

Terminal pins	Signal
TB16-1	Utility CB feedback
TB16-2	Genset CB feedback
TB16-3	Utility CB CLOSE command
TB16-4	Utility CB OPEN command
TB16-5	Genset CB CLOSE command
TB16-6	Genset CB OPEN command
TB16-7	REMOTE_START
TB16-8	CUSTOMER_OP1
TB16-9	CUSTOMER_OP2
TB16-10	CUSTOMER_INPUT_1
TB16-11	ESTOP_1
TB16-12	NC

8.Meter Calibration

Alternator Voltage and Current adjustment:

The meter calibration is done when the genset is running to adjust the Alternator voltage and Alternator current as per the customer reference meter. Adjustment is allowed +/-10% of the meter reading value.

The display must be calibrated as per the procedure below:

1. To enter meter calibration, press “Reset” and “OK” for 5 seconds when genset is in running state.
2. A screen will appear saying “Entering Calibration Mode”
3. Some of the pushbuttons on the operator panel are used for navigation as indicated by an arrow on the pushbutton.
4. When in meter calibration, the voltage screen will be displayed first as below.

CALIB MODE(1/3)		
	Vrms	Gain
VL12	999	1.000
VL23	999	1.000
VL31	999	1.000
←	↑	↓
		✓

CALIB MODE(2/3)		
	Irms	Gain
I1	99.9	1.000
I2	99.9	1.000
I3	99.9	1.000
←	↑	↓
		✓

5. Increase/Decrease the Gain value by pressing the ↑/ ↓ till the desired value is achieved for respective voltage or current values.
6. Press → (OK) to OK this setting and move to next voltage / current gain adjustments.
7. Once all settings are done press Reset to save the adjustments.
8. In case if the Genset stops while doing the adjustments the control will come out of the calibration mode and will not save the changes done.

Utility Voltage Adjustment:

The Utility calibration is done when the genset is Stop state to adjust the Utility voltage as per the customer reference meter. Adjustment is allowed +/-10% of the meter reading value.

The display must be calibrated as per the procedure below -

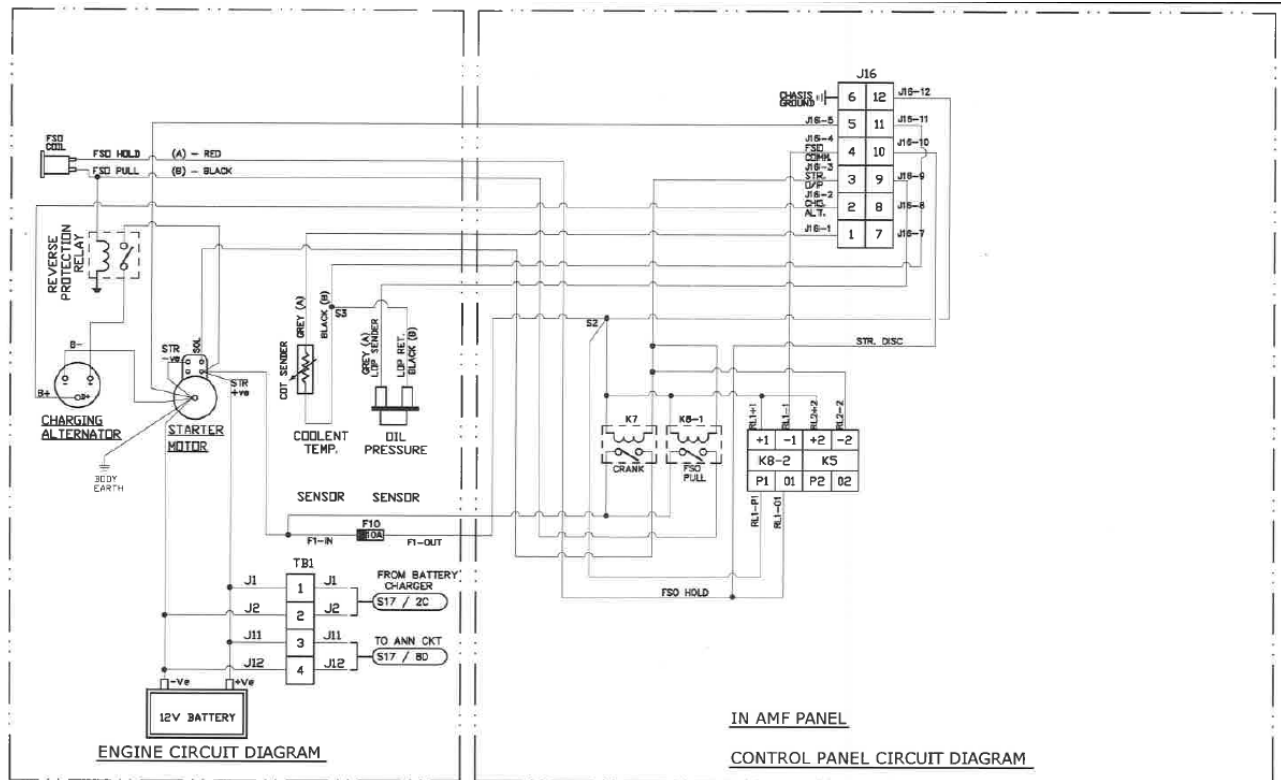
1. To enter meter calibration, press “Reset” and “OK” for 5 seconds when genset is in Stop state.
2. A screen will appear saying “Entering Configuration Mode”
3. Some of the pushbuttons on the operator panel are used for navigation as indicated by an arrow on the pushbutton.
4. Navigate to “AMF SETUP”.
5. When in meter calibration, the voltage screen will be displayed as below.

AMF SETUP(8/8)			
	Vrms	Gain	
VL12	999	1.100	
VL23	999	1.100	
VL31	999	1.100	
←	↑	↓	✓

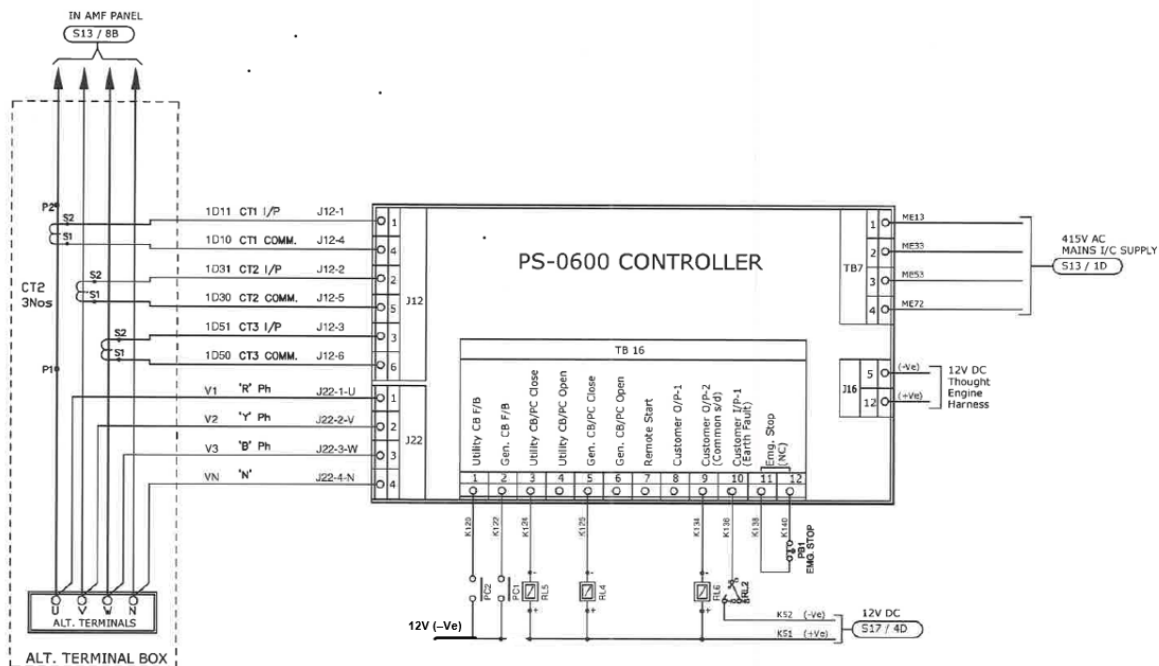
6. Increase/Decrease the Gain value by pressing the ↑/ ↓ till the desired value is achieved for respective voltage or current values.
7. Press → (OK) to OK this setting and move to next screens of AMF SETUP.
8. Once all settings are done press OK to save the adjustments.

9. Wiring Diagram

Engine Circuit diagram:



Alternator circuit diagram:



10.Commissioning Report

Dealer promotion ID	
Service Engineer	
Dealer Name	
Engine Model	
Alternator	
Serial No	
Controller Serial No	
GSN	
Engine Running Hours	
Commissioning Hours	
Control Run Time	
KW Hours -1) Genset 2) Utility	
RPM	
Lube Oil Pressure	
Coolant Temperature	
O/P Voltage	
Current	
KW	
RTC	
Actual Date & Time	

In case of Doubts don't hesitate to contact the nearest Head Zonal Electronics.

Zone	Name	Mobile
North	Pradeep Nagar	9891799305
East	Ritesh Ranjan	7604047971
West	Rajendra Salokhe	9011027355
South	Konnur BS	9448496780